Unit 1

1. Introduction to AI

Module 1.8

1.8 Pre-Trained Models

Tutorial 1.8.3

1.8.3 Risks of using Pre-Trained Models and Transfer Learning

AI-1.8.1 K2 Explain the use of pre-trained AI models and the risks associated with them.

Risks of using Pre-Trained Models and Transfer Learnin

Using pre-trained models and transfer learning are both common approaches to building AI-based systems, but there are some risks associated which include:

- **Lack of transparency** - A pre-trained model may lack transparency compared to an internally generated model.
- **Domain Mismatch** - The level of similarity between the function performed by the pre-trained model and the required functionality may be insufficient.
- **Likelihood of unknown defects** - If the pre-trained model is not widely used, there are likely to be more unknown defects and more rigorous testing may be needed to mitigate this risk.
- **Inherited biases** - may not be apparent if there is a lack of documentation about the data used to train the model.

Risks of using Pre-Trained Models and Transfer Learning Cont

- **Data Privacy Concerns** - Pre-Trained models may have used proprietary data which may raise ethical or legal issues.
- **Security Vulnerabilities** - Models created through transfer learning are highly likely to be sensitive to the same vulnerabilities as the pre-trained model on which it is based.
- **Version Drift** - Updates to pre-trained models can change behavior which might break downstream dependencies.
- **Lack of Explainability** - Many pre-trained models are black boxes, making it hard to interpret predictions or debug errors.

Note that several of the above risks can be more easily mitigated by having thorough documentation available for the pre-trained model

THANK YOU!!!